$R-NH_2$ ⇒ Primary / 1° Amine
$R-NH-R$ ⇒ Secondary / 2° Amine
$R-\underset{R}{N}-R$ ⇒ Tertiary / 3° Amine
} All are treated as diff. functional groups

$\left(R-\overset{R}{\underset{R}{\overset{|}{\underset{|}{N^+}}}}-R\right)X^-$ ⇒ Quaternary Ammonium Salt

Method of Preparation of Amines :

1) Ammonolysis of alkyl halide :→

* With every step, the electrophilicity of nitrogen keeps increasing, hence instead of Amine, it gives Salt.

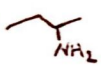
$R-X + NH_3 \longrightarrow (R-\overset{+}{N}H_3)X^- \longrightarrow R-NH_2$

$R-X \rightarrow R-NH-R \xrightarrow{R-X} R-\underset{R}{N}-R \xrightarrow{RX} \left(R-\overset{R}{\underset{R}{\overset{|}{\underset{|}{N^+}}}}-R\right)X^-$

* In order to form Amine, we need to take R-X in excess & NH_3 in limited amount.

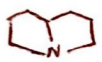
$R-X + NH_3 \longrightarrow R-NH_2$
(Excess)

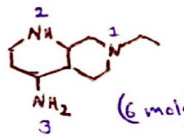
Ques How many moles of RX will be req. to convert following into quaternary salts :

1)  (3 mol.)

2) CH_3-NH_2 (3 mol.)

3)  (2 mol.)

4)  (1 mol)

5)  (6 mol.)

2) Reduction of Nitro Compounds :→

$R/Ar-NO_2 \xrightarrow{P} R/Ar-NH_2$

* Here, P can be : 1) H_2/Pd, EtOH
2) Sn / HCl
3) Fe / HCl
4) $LiAlH_4$ (LAH)

Red^n Pattern ↓

$NO_2 \rightarrow N=O \rightarrow NH-OH \rightarrow NH_2$

* In order to form amine, 3 mol. Sn & 6 mol. HCl is consumed, while to go till salt, 7 mol. HCl is consumed.

Eg $R/Ar-NO_2 \xrightarrow[6\,HCl]{3\,Sn} R/Ar-NH_2 \xrightarrow{HCl} (R/Ar-\overset{+}{N}H_3)Cl^-$

3) Reduction of Nitriles :→

$R-C\equiv N \xrightarrow{P} R-CH_2-NH_2$

* P can be : 1) Sn/HCl or Fe/HCl
2) LAH
3) Na/EtOH

Amines

4) H_2/Pd 6) B_2H_6

5) (i) $SnCl_2/HCl$ (ii) $NaBH_4$
(Stephen's Reduction Then Red^n by SBH)

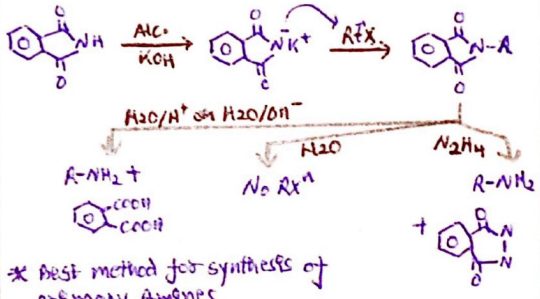
Eg : $R-NC \xrightarrow{H_2/Pd} R-NH-CH_3$
(2° Amine)

4) Reduction of Amide :→

$Ar/R-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow[3)\ H_2/Pd]{1)\ LAH \quad 2)\ B_2H_6} Ar/R-CH_2-NH_2$

Any one of these 3 Reducing agents can be used.

5) Gabriel Phthalimide Synthesis :→



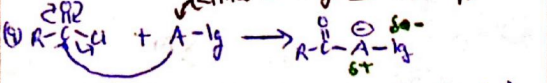
* Best method for synthesis of primary Amines
* Second step is S_N2 i.e. Inversion occurs.
* In R-X, 'R' cannot be the group which does not support S_N2 or inversion like ( , 3° Alkyl Halide,  , etc)

Ques Which of the following can be formed by Gabriel Phthalimide Synthesis ?

1) Yes
2) CH_3-NH_2 Yes
3)  No
4)  No
5) H_2N-CH_2-COOH Yes
6)  No
7) $(CH_3)_3N$ No
8)  No

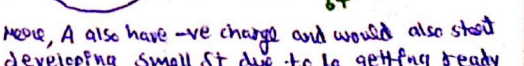
6) Hoffmann Bromamide Degradation Rx^n :→

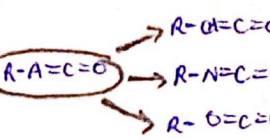
Wolff Rearrangement ⇒

(i) 
(Either 2- charge on 1- & 1p)

Here, A also have -ve charge and would also start developing small +ve due to lg getting ready to leave.

If, A = C → Carbene like character
A = N → Nitrene like character
A = O → Nascent oxygen like character

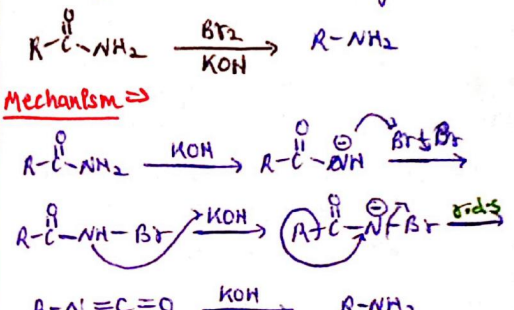
(ii) 



* Similar kind of rearrangement takes place in Hoffmann Bromamide Degradation.

$R-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow[KOH]{Br_2} R-NH_2$

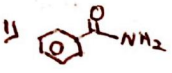
Mechanism ⇒

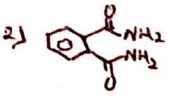


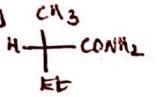
$R-N=C=O$ (Isocyanate) $\xrightarrow[(H_2O)\ Hydrolysis]{KOH} R-NH_2$

* Isocyanate is the only intermediate for this rx^n
* Methyl Isocyanate (Me-NCO) is a toxic gas (Bhopal gas tragedy)
* 4 mol KOH & 1 mol. Br_2 is consumed in the reaction.

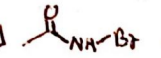
4 mol. KOH → 2 mol. KOH in Acid base
→ 2 mol. KOH in Hydrolysis

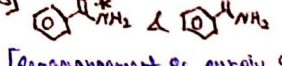
1) 

2) 

3) 

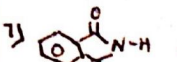
During Rearrangement configuration never changes.

4) 

5) 

[Rearrangement is purely intramolecular hence rearranging group won't exchange among two or more compounds]

6) $CH_3-COOH \xrightarrow[(ii)\ A]{(i)\ NH_3} \xrightarrow[(iii)\ Br_2/KOD]{} CH_3-ND_2$

7) 

7) Curtius Rearrangement (Basic Medium) :→

$R-\overset{O}{\overset{\|}{C}}-Cl \xrightarrow[(ii)\ H_2O/OH^-]{(i)\ NaN_3} R-NH_2$

Properties of Amine :

1) Basic Strength :→

In Gaseous Phase ⇒

Here, only inductive effect is the deciding factor for basic strength.

Order : 3° Amine > 2° Amine > 1° Amine > NH_3

In Aqueous Phase ⇒

Here, there are two deciding factor i.e Solvation in water and inductive effect.

→ So, due to solvation, 2° Amine is most basic and depending upon 'R', basicity of 1° & 3° Amine is decided.

In Aqueous Phase

R = Me ⇒ $2^\circ > 1^\circ > 3^\circ$

R = Et ⇒ $2^\circ > 3^\circ > 1^\circ$

2) Acylation :→

$R-\ddot{N}H_2 + CH_3-\overset{O}{\overset{\|}{C}}-Cl \xrightarrow[\text{or } NEt_3]{Py} R-\overset{O}{\overset{\|}{C}}-NH-R + HCl$

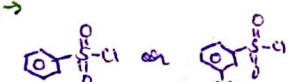
Here bulky base like Pyridine or $N(Et)_3$ are taken so that HCl get consumed & rxn moves forward.

$R-OH + CH_3-\overset{O}{\overset{\|}{C}}-Cl \xrightarrow{Pyridine} R-O-\overset{O}{\overset{\|}{C}}-CH_3$

(Change in mass = 42g)

Means if mass of product rises by n(42) g, then compound must have had 'n' number of free -OH groups.

3) Hinsberg's Test :→

Hinsberg Reagent ⇒ 

(Tosyl chloride)

* Test used for separation of 1°, 2° and 3° Amines.

$R-NH_2 + Ph-SO_2-Cl \xrightarrow{Py} Ph-SO_2-NH-R \xrightarrow{Alkali}$ Soluble

$R-NH-R + Ph-SO_2-Cl \xrightarrow{Py} Ph-SO_2-N(R)-R \xrightarrow{Alkali}$ Insoluble

$R-\underset{R}{\overset{R}{N}}-R + Ph-SO_2-Cl \xrightarrow{Py}$ No Rxn

4) Carbyl Amine Test :→

$R-NH_2 \xrightarrow[3KOH]{CHCl_3} R-NC$

* Foul smell
* Isocyanide
* Only 1° amine give rxn

Mechanism ⇒

(i) $H-C(Cl)(Cl)-Cl + KOH \longrightarrow K\overset{+}{C}Cl_3 \xrightarrow{-H_2O}$ (Dichloro carbene) $:CCl_2 + KCl$

(ii) $R-\ddot{N}H_2 + :CCl_2 \xrightarrow{rds} R-\overset{\oplus}{N}H-\overset{\ominus}{C}Cl_2 \xrightarrow{-HCl} R-\ddot{N}=\ddot{C}-Cl \xrightarrow{-HCl} R-N\equiv C$

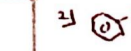
3KOH → 1 KOH for DCC formation
→ 2 KOH, to consume HCl produced

→ If yc characteristic rxn of 1° Amines

→ ROR ↓ses upon add^n of more KCl due to common ion effect.

Que Which of the following would give carbyl amine Test.

1) CH_3-NH_2 ✓ 2) (2°) $-NH-$ ✗ 5) $N(CH_3)_3$ ✗ (3°)

3) ✓ 3)  ✗ (Amide)

5) Reaction with Nitrous Acid (Diazotisation) :→

$R-NH_2 \xrightarrow[\text{or } HNO_2]{NaNO_2/HCl} R-N_2^+ \xrightarrow{-N_2} R^\oplus$ → $R-OH$, $R-Cl$, $R-NO_2$, $R-ONO$, Alkene

* There are many Nu^- in the system that would attack on R^+ and form product.
* But, major is Alcohol (R-OH)

Important Points ⇒

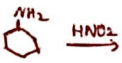
(i) HNO_2 behaves as a base here

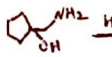
(ii) Carbocation Rearrangement possible

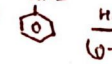
(iii) $\overset{\oplus}{N}=O$ (Nitrosonium ion) is attacking electrophile

(iv) R-OH is major ⇒ When $NaNO_2/HCl$ (HNO_2) taken
R-Cl is major ⇒ When 'NOCl' taken (Tilden Reagent)

(v) 2° Amine forms 'Yellow oily liquid' with HNO_2

1) 

2) 

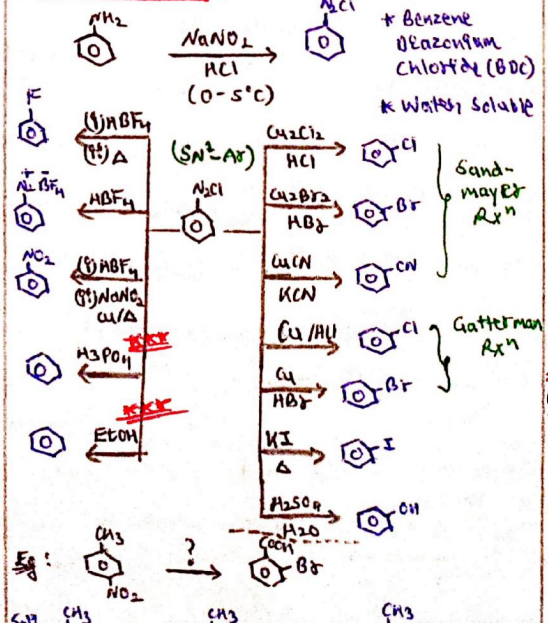
3) 

4) $CH_3-NH_2 \xrightarrow{\Delta} CH_3-OH$

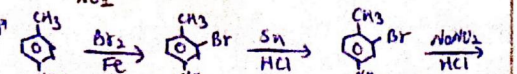
5) $R_2-(NH) \xrightarrow{\Delta} R_2-N-N=O$ (Yellow oily liq.)

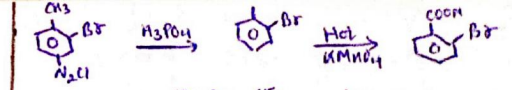
6) $R_3N \xrightarrow{\Delta}$ No Rxn

Diazonium Salt :

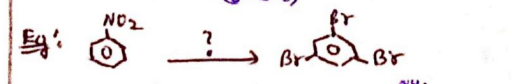


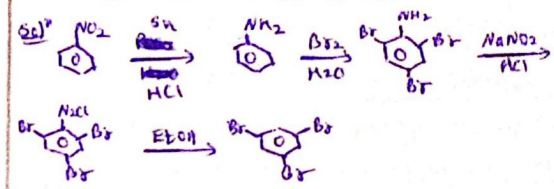
Eg :

Sol^n 



Final Ans ⇒ (i) Br_2/Fe (ii) Sn/HCl (iii) $NaNO_2/HCl$ (0-5°C) (iv) H_3PO_4 (v) $\Delta, KMnO_4$

Eg : 

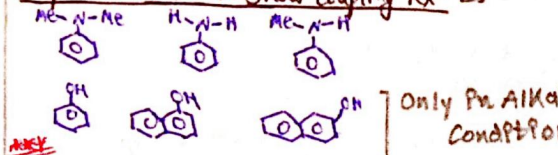
Sol^n 

Coupling Reaction :→

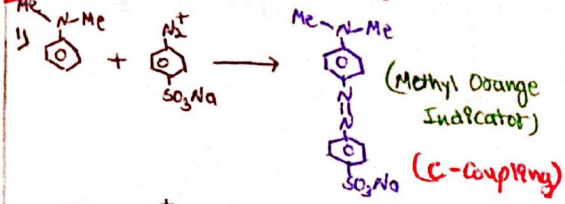
→ $Ar-N_2Cl$ reacts only with highly activated arys because it itself is a weak electrophile.

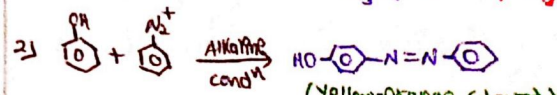
→ Medium during rxn is kept slightly basic in case of -OH, in order to form $-O^-$ & rise e^- density in the ring.

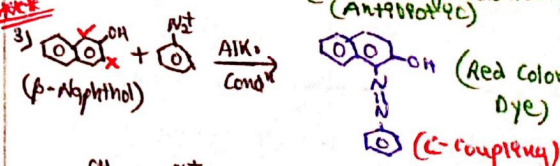
Compounds that can show coupling Rxn ⇒

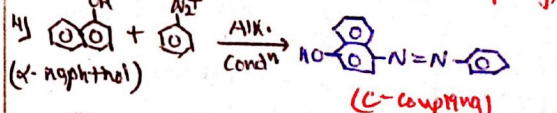


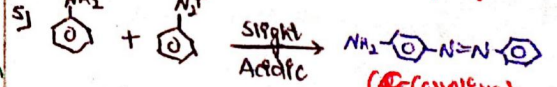
Only in Alkaline Condition

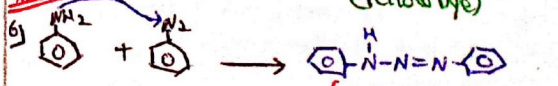
1) 
(Methyl Orange Indicator)
(C-coupling)

2) 
(Yellow-Orange Coloured dye)
(Antipyretic)

3) 
(β-Naphthol) (Red Colour Dye)
(C-coupling)

4) 
(α-naphthol)
(C-coupling)

5) 
(C-coupling)
(Yellow Dye)

6) 
(N-coupling)

In normal conditions, w/o acidic medium, N-N coupling takes place in case of 1° amine